=) EMI :- Calc monthly emi for loan

$$\therefore EMI = \frac{P \times r \times (r+1)^n}{(r+1)^n - 1}$$

Where :-

P =) Principal
r =) Monthly interest rate
n =) Loan Tenure (Monthly)

=)

=) FOIR (Fixed Obligation to Income Ratio)

$$\therefore FOIR = \frac{\text{Existing EMI} + \text{Other oblig.} + \text{New EMI}}{\text{Monthly Income}} \times 100$$

Below 40% → Good
40 - 50% → Moderate
Above 50% → High Risk

=) DTI (Debt To Income Ratio)

$$\therefore DTI = \frac{\text{Current Debt}}{\text{Monthly Income}} \times 100$$

∴ Lower is better.

1) **Max Principal :-** Inverse of the EMI formula

$$P = E \times \left[\frac{(1+r)^n - 1}{r(1+r)^n} \right]$$

where :-

P → Max Principal (Loan Amt)
E → Target EMI
r → Monthly interest rate = $\frac{\text{Annual Interest Rate}}{12 \times 100}$

n → Total loan tenure in months

2) **Saving Ratio :-**

$$\text{saving ratio (\%)} = \left(\frac{\text{Monthly income} - \text{Monthly Exp.}}{\text{Monthly Income}} \right) \times 100$$

3) **Emergency fund Coverage :-**

•

$$\text{Emer. f. Cov. (Monthly)} = \frac{\text{Total Emergency Savings}}{\text{Monthly Living expenses}}$$

(4) Disposable Income :

$$D.I = \text{Monthly inc} - \text{Monthly EMI} - \text{other obligation}$$

Score :

$$\text{Ratio} = \frac{\text{Disposable Income}}{\text{Monthly Income}}$$

→ Max ratio can't be >= 0.40 [Max score = ratio >= 40%]

(5) Credit Utilization Score :-

$$\frac{\text{Credit used}}{\text{Credit limit}} \times 100$$

❀ Weight of each metrics :-

(1) ~~FOIR (lower is better)~~

~~<= 30%~~	~~=~~	~~100~~
~~30 - 40%~~	~~=~~	~~80~~
~~40 - 50%~~	~~=~~	~~60~~
~~50 - 60%~~	~~=~~	~~40~~
~~> 60%~~	~~=~~	~~20~~

Good Score : 80 - 100

(2) DTI (Lower is better)

< = 20 %	=	100
20 - 35 %	=	80
35 - 50 %	=	60
50 - 60 %	=	40
> 60 %	=	20

G.S = 90 - 100

(3) Credit Score (Based on CIBIL)

>= 750	=	100
700 - 749	=	80
650 - 699	=	60
600 - 649	=	40
< 600	=	20
Null	=	50

Penalty :

Existing loan > 2 then

Penalty = (loans - 2) × 5

Exp :

cibil = 760, loans = 4

P = (4 - 2) × 5 = 10

100 - 10 = 90

Good Score 90 - 10

(4) <u>Saving Ratio</u>

High	=	100
Good	=	80
Moderate	=	60
Low	=	40
Very Low	=	20

(5) <u>Employment Stability Score</u> :-

Experience :

>= 5	years	=	40
2 - 4	"	=	25
< 2	"	=	10

+

Income Stability :

Stable	=	60
Moderate	=	35
Unstable	=	15
Null	=	35

(6) ~~Emergency Fund~~

~~6 + Months = 100~~
~~4 - 6 " = 80~~
~~2 - 4 " = 60~~
~~1 - 2 " = 40~~
~~< 1 " = 20~~

(7) Credit Score Utilization

< 30% = 100
30 - 50% = 75
50 - 75% = 50
> 75% = 25

∴ Lower = better

(8) Disposable Income Score

Ratio ~~[illegible]~~ $= \dfrac{D.I}{Income}$

$Score\ (100) = \dfrac{Ratio}{0.40} \times 100$

Max score = 100 if ratio is >= 40%

>= 40 %	=	100
30 %	=	75
20 %	=	50
10 %	=	25
0 %	=	0

(1) Foir Score : It is loan-type dependent

Home loan, car loan, etc has different thresholds

General Scoring :-

<= Excellent limit	=	100
<= Acceptable limit	=	75
<= Caution limit	=	50
Above Caution	=	25

Example for Personal loan :

<= 40 %	=	100
40 - 50 %	=	75
50 - 60 %	=	50
> 60 %	=	25

(2) DTI : Also loan-type dependent

General scoring :-

Low risk	=	100
Moderate	=	75
High	=	50
very High	=	25

(4.1) Saving Ratio Score :

>= 30%	=	100
20 - 30%	=	75
10 - 20%	=	50
< 10%	=	25

(6) Emergency fund score :-

>= 6 Months	=	100
3 - 6 "	=	75
1 - 3 "	=	50
< 1 "	=	25

∴

Applying Weights :-

1) FOIR = 25 %
2) DTI = 20 %
3) Credit Profile = 20 %
4) Saving Ratio = 10 %
5) Employment Stability = 10 %
6) Emergency Fund = 5 %
7) Credit Utilization = 5 %
8) Disposable Income = 5 %

Final formula :-

∴ Financial Score = (foir × 0.25) + (DTI × 0.20)
+ (credit × 0.20) + (Savings × 0.10)
+ (Employment × 0.10)
+ (Emergency × 0.05)
+ (credit util. × 0.05)
+ (disposable × 0.05)

Risk level :-

90 - 100 = Excellent
75 - 89 = Good
60 - 74 = Moderate
40 - 59 = Needs Improvement
0 - 39 = High Risk